

EDUCATION

Bachelor of Computer Science (Cybersecurity)

Sept 2022 — Jul 2025

Asia Pacific University (APU), Bukit Jalil • CGPA: 3.81 • First Class Distinction

Penetration Testing • Network Design • Linux Administration • AWS Cloud • Deep Learning for Security • Web & Mobile Development

PROFESSIONAL EXPERIENCE

AI Security Engineer — Maybank HQ, Menara Maybank

Jan 2026 — Present

- Develop and implement AI-driven security solutions within the AI Security Engineering team
- Conduct offensive security assessments across web applications, APIs, and mobile platforms
- Active bug bounty researcher with confirmed vulnerabilities reported on Bugcrowd and GitHub open-source repositories
- Discovered and responsibly disclosed 6 CVEs across open-source projects (HeyForm, Open WebUI, compliance-trestle)

Cybersecurity Intern — Maybank HQ, Menara Maybank

Aug 2024 — Nov 2024

- Performed internal penetration testing on web, mobile, API, infrastructure, and NAC systems
- Discovered critical vulnerabilities in payment APIs, OTP flows, and wireless networks
- Delivered security reports aligned with Bank Negara Malaysia (BNM) RMIT compliance standards
- Strengthened internal cyber resilience through systematic vulnerability assessment and exploit validation

Cloud Trainee — PayNet, AWS re/Start Program

Jun 2024 — Jul 2024

- Trained in AWS compute, storage, networking, and security fundamentals
- Deployed cloud services (EC2, IAM, RDS, S3) and configured monitoring with CloudWatch
- Earned AWS Certified Cloud Practitioner credential

AWARDS & ACHIEVEMENTS

AWARD	EVENT	YEAR
Champion + Best Demo	Sparkathon 2025 (APU x BAT) — JetCycle AI Tech Swap Station	2025
Champion — Security	NexG Godam!ah! 2.0 Smart ID Hackathon — MyLayak Zero-Trust Platform	2025
Gold Award	MIJC2026 Virtual Innovation — eKYC Biometric Authentication Research	2026
Top 15 / 100+ Teams	EY Young Technology Professional Challenge (YTPC) 2025 — Trendify	2025
Blueprint Finalist	LUNA Women's Safety App & Wira Disaster Resilience Platform	2026

SECURITY RESEARCH (CVEs)

SEVERITY	VULNERABILITY	PROJECT	ID
HIGH	Stored XSS via Unauthenticated SVG File Upload	heyform/heyform	CVE-2026-45797
HIGH	SSRF via Unauthenticated Image Proxy	heyform/heyform	GHSA-3j2g-3q97-fvv5
HIGH	Arbitrary File Write via Cache Path Traversal	oscal-compass/compliance-trestle	GHSA-mj4x-vf5c-5xg8
HIGH	Arbitrary File Write via Cache Path Traversal	oscal-compass/compliance-trestle	GHSA-g3vg-vx23-3858
MODERATE	Critical SSRF in Remote Fetching Subsystem	oscal-compass/compliance-trestle	GHSA-w76h-q7c6-jpjp
MODERATE	Mass Assignment — User ID Spoofing	open-webui/open-webui	CVE-2026-45396

KEY PROJECTS

MyLayak — Zero-Trust Eligibility Platform

Champion, Godam!ah! 2.0

React, TypeScript, Node.js, HMAC-SHA256, XOR+ROT13 Obfuscation, QKD (BB84)

- Architected a zero-trust eligibility platform allowing citizens to discover government subsidies without exposing personal data
- Implemented dual-layer steganographic QR authentication on MyDigital ID to prevent QR faking and copying
- Designed Soulbound Eligibility Tokens (SETs) with server-side JWT issuance, pattern obfuscation, and BB84 quantum key distribution

eKYC Biometric Authentication System — Final Year Project

Gold, MIJC2026

React, TypeScript, Tailwind, Python ML, AES-256, Supabase RLS, REST APIs

- Designed a multi-modal biometric framework for AI-resilient digital identity verification
- Integrated facial recognition, liveness detection (EAR/MAR + landmark dynamics), and behavioural biometrics for continuous post-login authentication
- Engineered anomaly detection engine to combat deepfakes, synthetic identities, and session hijacking

LUNA — Women's Safety App

Blueprint Finalist

React, TypeScript, Supabase, Firebase, Google Gemini Vision, Azure Cognitive Services

- Built a comprehensive safety application disguised as a period tracker with encrypted evidence vault and AI forensic analysis
- Implemented deepfake photo immunization using adversarial perturbation and steganographic verification (SHA-256)
- Integrated Google Gemini AI for evidence analysis, Azure Neural TTS, and real-time boundary-setting coaching

KongsiGo — TNG Group Wallet (TNGD FinHack)

Hackathon

Python (FastAPI), React 18, TypeScript, PostgreSQL, Redis, WebSockets, ONNX ML, Claude/Ollama LLMs, AWS Lambda

- Built a shared-wallet engine on Touch 'n Go eWallet with Trip and Family pools using atomic, double-entry ledgering
- Implemented device-bound Phone + PIN auth and HMAC-sealed passwordless login with AWS Lambda verification
- Engineered democratic spend voting (majority/unanimous/threshold/admin) with early reject and emergency override
- Built on-device ML (DistilBERT tx classifier + Isolation Forest anomaly detector) and dual AI agents for fraud detection

Wira — AI Disaster Resilience Platform

Blueprint Finalist

React, TypeScript, Vite, Tailwind CSS, Recharts, Google Maps API

- Developed gamified disaster preparedness platform for ASEAN covering 10 countries with real disaster data
- Built Possum Protocol — offline emergency toolkit with mesh networking, GPS sharing, and distress signal detection

JetCycle — AI Tech Swap Station

Champion + Best Demo, Sparkathon 2025

- Designed an AI-powered smart kiosk with robotic arm for automated e-waste dismantling and component categorization
- Created full product workflow from device scanning to digital reward distribution

SipMate — AI Wine Discovery App

World's Largest Hackathon (Bolt)

React Native, Expo, Supabase, DeepSeek API

- Built a wine discovery app with 3,000+ wines and AI-generated descriptions using DeepSeek API
- Developed personalized recommendation engine; planning RAG integration for improved accuracy

CERTIFICATIONS

AWS Certified Cloud Practitioner (2024)

Cisco NetAcad — Switching & Routing Essentials (Merit)

EXTRACURRICULAR

- Bug Bounty Researcher — Active on Bugcrowd; findings on GitHub OSS repos
- CTF Competitions — rEntas CTF, IWD CTF, picoCTF 2024
- Solana Hackathon — Blockchain trading platform with Web3 tools
- Google Hackathon — Foodie Map, location-based food discovery

TECHNICAL SKILLS

SECURITY	Penetration Testing, Web/API/Mobile Security, Bug Bounty, Zero Trust Architecture, AI/ML Security
LANGUAGES	Python, TypeScript, Java, SQL, JavaScript, C#
FRAMEWORKS	React, React Native, Tailwind CSS, Vite, Supabase, Firebase
TOOLS	Burp Suite, Wireshark, Frida, Nmap, Git
CLOUD	AWS (EC2, IAM, RDS, S3, CloudWatch), Supabase, Firebase Hosting
SOFT SKILLS	Technical Reporting, Team Leadership, Problem Solving, Critical Thinking